GenAI4 For Teleoperation video prediction.

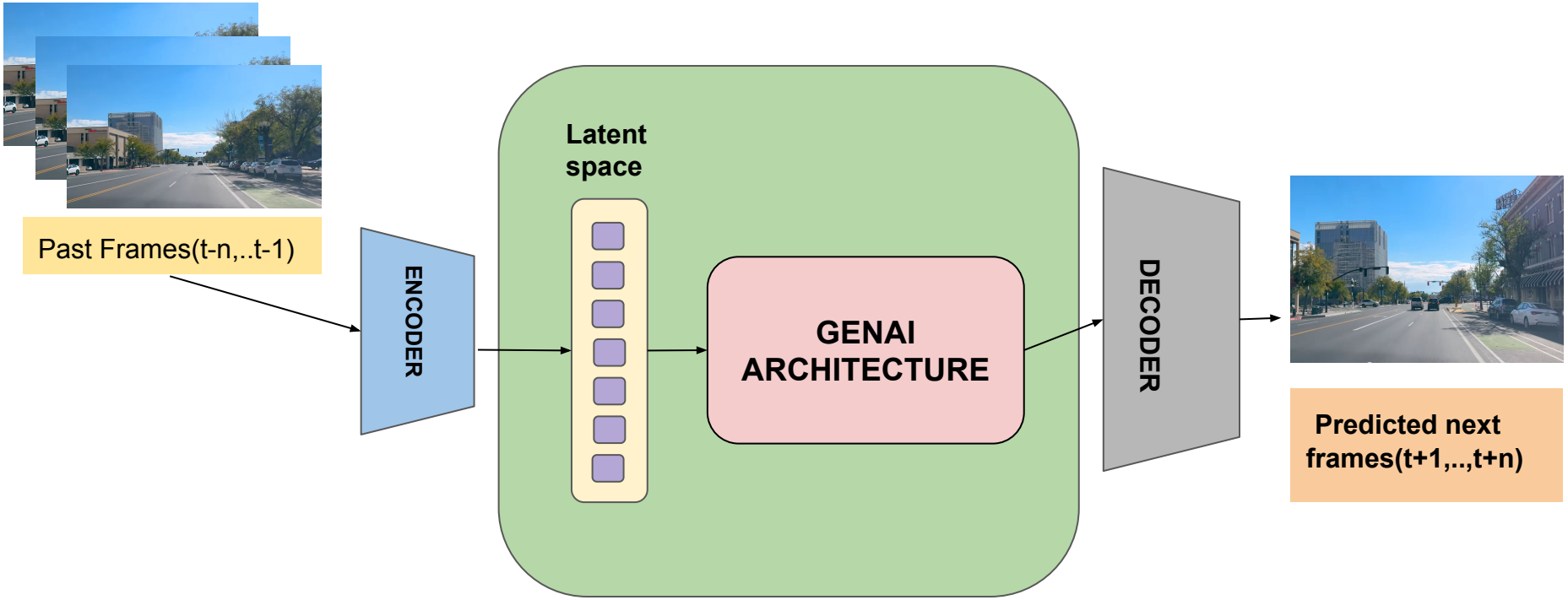
Original video

Dimensions	1024 × 768
Video Codec	MPEG-4 Video (Simple Profile)
Video Bit Rate	2427 kbps
Frame Rate	15.00 frames per second

MagVIT2

Dimensions	128 × 128
Video Codec	MPEG-4 Video (Simple Profile)
Video Bit Rate	387 kbps
Frame Rate	24.00 frames per second

VIDEO IN - ENCODE - PREDICT - DECODE- VIDEO OUT



How the MAGVIT2 WORKS

Step 1 — Video Loading & Preprocessing

Original Video:
cam2_front.mp4 (e.g., 1920×1080)

Transforms Applied:

1. Resize height → **128 px** (keep aspect ratio)
2. Center crop → **128×128**

Result:
`[17 frames × 128 × 128 × 3] = 835,584 values`

Visualization:
Original → ↓ Resize → ↓ Crop → ↓ Normalize → → 128×128×17 tensor

Step 2 — 3D CNN Encoder

- downsampling
- Expands channels (3 → 18)
- Encodes structure, motion, and color

Output: `[1, 18, 5, 16, 16]`
(Compressed latent video representation)

Step 3 — Lookup-Free Quantization (LFQ)

- Maps each 18-D latent vector to nearest token in **262,144-entry codebook**
- Full codebook utilization (no dead codes)
- Each spatial-temporal position → 1 token

Total tokens: `16×16×5 = 1,280`
→ Video becomes a **sequence of 1,280 discrete tokens**

Step 4 — Autoregressive Transformer (MAGVIT2 Core)

Input: sequence of past tokens
Goal: predict the probability distribution of the next token

Architecture:

- I used GPT-2 **decoder-only transformer since no pretrained was available**
- **Causal attention** (no future information)
- Can condition on:
  - Previous tokens
  - Context (text, actions, camera views)

Step 5 — Video Generation Process

Generation = Autoregressive Sampling

1. Start with context tokens (e.g., first frames or prompt)
2. Transformer predicts next token
3. Sample → append → feed back into model
4. Repeat until 1,280 tokens generated
5. → Then decode generated tokens into video frames

Step 6 — Decoding (Post-Generation)

Input: predicted token sequence
Process:

- Lookup tokens in LFQ codebook → latent space `[1,18,5,16,16]`
- 3D Transposed CNN upsamples
- Output reconstructed video `[1,3,17,128,128]`

Epona: Autoregressive Diffusion World Model for Autonomous Driving

Dimensions	1024 × 512
Video Codec	MPEG-4 Video (Simple Profile)
Video Bit Rate	584 kbps
Frame Rate	5.00 frames per second

Distribution Shift in Autoregressive Generation

```
┌──────────────────────────────────────────────────────────────────────┐
│ TRAINING vs INFERENCE MISMATCH                                       │
├──────────────────────────────────────────────────────────────────────┤
│                                                                      │
│ During Training:                                                     │
│                                                                      │
│ ┌──────────┐   ┌──────────┐   ┌──────────┐                          │
│ │ Real     │ → │ Real     │ → │ Real     │                          │
│ │ Frame 1  │   │ Frame 2  │   │ Frame 3  │                          │
│ └──────────┘   └──────────┘   └──────────┘                          │
│                                                                      │
│   Model learns: Real → Real                                          │
│                                                                      │
│ During Inference (Autoregressive):                                   │
│                                                                      │
│ ┌──────────┐   ┌──────────┐   ┌──────────┐                          │
│ │ Real     │ → │ Predicted│ → │ Predicted│                          │
│ │ Frame 1  │   │ Frame 2  │   │ Frame 3  │                          │
│ └──────────┘   └──────────┘   └──────────┘                          │
│                                                                      │
│   Model sees: Predicted → Predicted                                  │
│                                                                      │
│ Result: DISTRIBUTION DRIFT                                           │
│ Quality: 100% → 98% → 95% → 90% → ... → 40%                          │
└──────────────────────────────────────────────────────────────────────┘
```

SLIDING WINDOW ALGORITHM

Initialize Window
(10 real frames)

PREDICTION PHASE
Generate 3 frames
(0.6 seconds ahead)

Update Window
(slide by 3)

INJECTION PHASE
Add 5 new real frames
🔄 Reset distribution

More frames needed? —Yes→

No

Output 30s video
64% real, 36% pred

Pure Autoregressive vs Sliding Window

Aspect	Pure Autoregressive	Sliding Window (Ours)
Window Composition	100% → 0% real	Always 50% real
Prediction Horizon	Unlimited	Short (3 frames = 0.6s)
Real Data Injection	Only at start	Every 8 frames
Quality at 30s	40-60%	95-98%
Distribution Drift	High ❌	None ✅
Realistic for AV	No	Yes ✅

Sliding Window Method

Metric	Value	
Total frames	150	
Predicted frames	54	
Real frames used	96 (10 initial + 90 injected)	
Injection cycles	18	
Duration	30 seconds @ 5fps	
Real frame %	64% (96/150)	
Window composition	50% real at all times	

Framework for testing

Video Prediction Methods: Deterministic vs Stochastic

Deterministic Models

(IRFVP, DMVFN, FVS)

Key Points:

- Single Best Prediction
- Sharp & Fast
- Fixed Outcome

Strengths:

- ✅ High Speed
- ✅ Real-Time
- ✅ Sharp Frames

Weaknesses:

- ❌ No Multiple Futures
- ❌ Poor Long-Term Accuracy

VS

Stochastic Models

(ExtDM, RAMVid, MCVD)

Key Points:

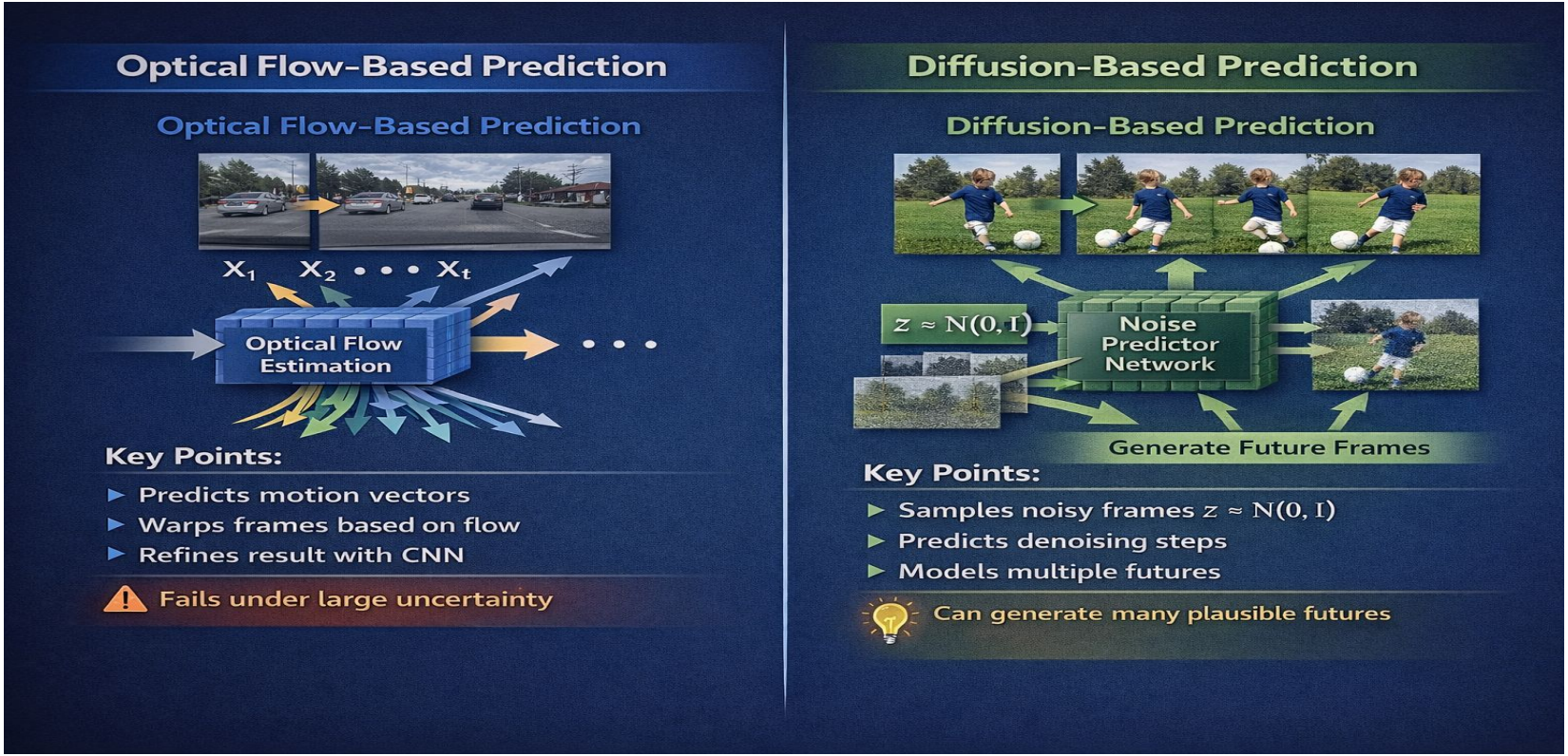
- Multiple Possible Futures
- Uncertainty Modeling
- Diverse Outcomes

Strengths:

- ✅ Multi-Modal
- ✅ Long Horizon
- ✅ Captures Variability

Weaknesses:

- ❌ Slow & Costly
- ❌ Hard to Deploy



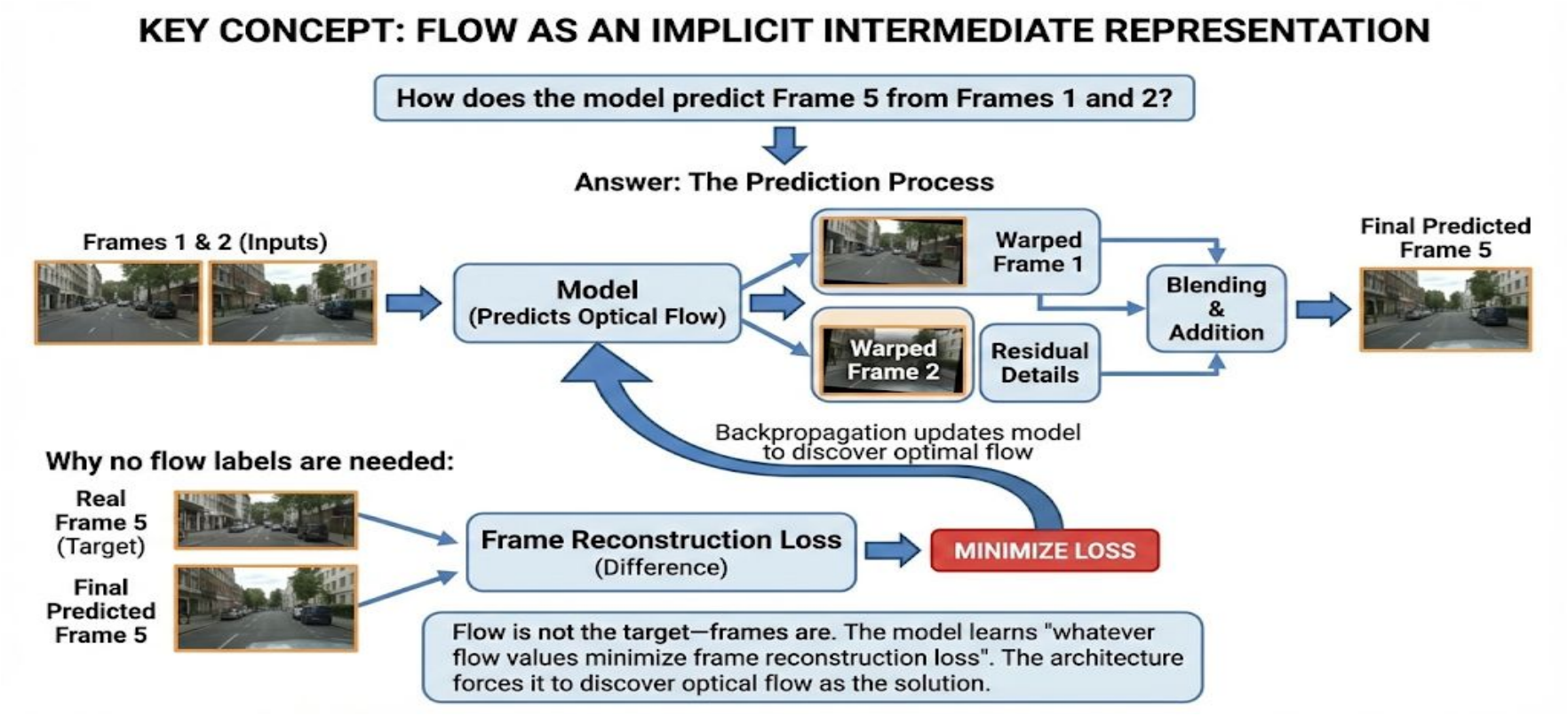
For AV teleoperation, optical-flow / deterministic video prediction is the better choice because it delivers **low latency, stable, single-future outputs** that humans can react to in real time.
Diffusion-based models are a poor fit here: their sampling latency and multi-future uncertainty add delay and visual instability, which actively hurts human decision-making during teleop

Pipeline of Extdm:

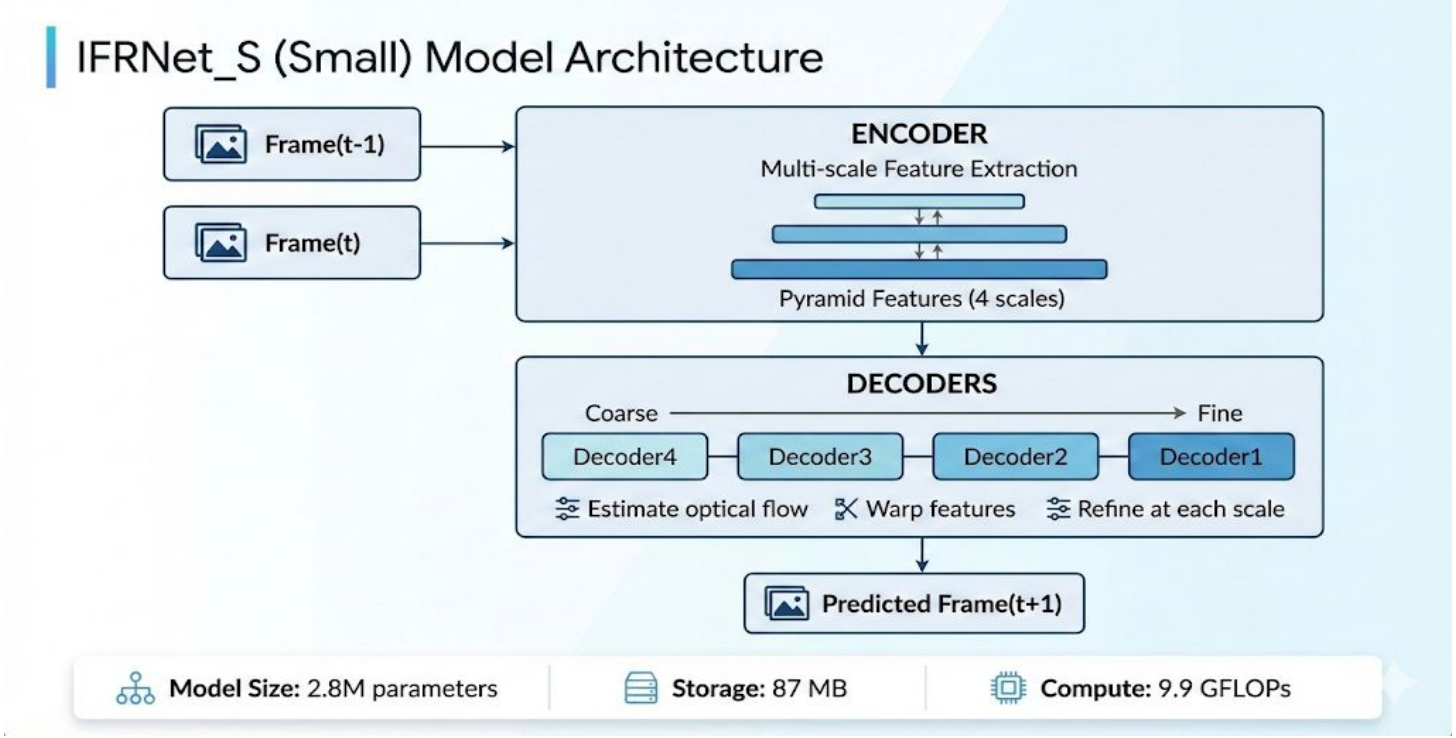
1. AV captures frame **100**
2. Due to latency, receiver currently has frame **97**
3. Receiver side predicts frame **100** using ExtDM
4. Receiver sends predicted frame **100** back to AV
5. AV already has **GT frame 100**
6. AV compares **100th vs 100 (GT)** → computes delta (non-learning)
7. AV sends delta back to receiver
8. Receiver uses delta as **conditioning** to generate frame **103 or 105** “better” using extdm
9. Receiver sees predicted video continuously
10. Every ~10 frames, delta is used to “correct accumulated error”

Optical Flow Estimation in IFRVP

Optical flow = per-pixel 2D motion between two consecutive frames.
Each pixel gets a vector (u, v) → how far it moved in x and y.



REAL-TIME VIDEO PREDICTION WITH FAST VIDEO INTERPOLATION MODEL AND PREDICTION TRAINING



Dataset Used:- BDD100K (https://bair.berkeley.edu/blog/2018/05/30/bdd/)

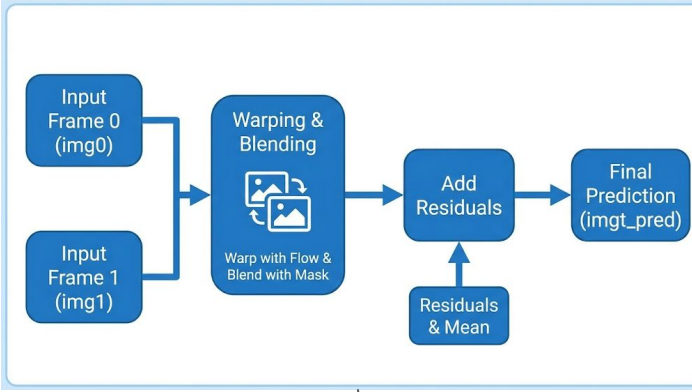
Frame work for independent model

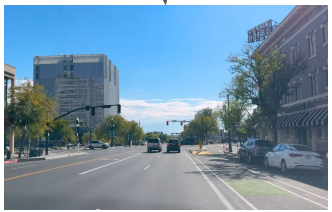
Past Frames(1,2)

ENCODER

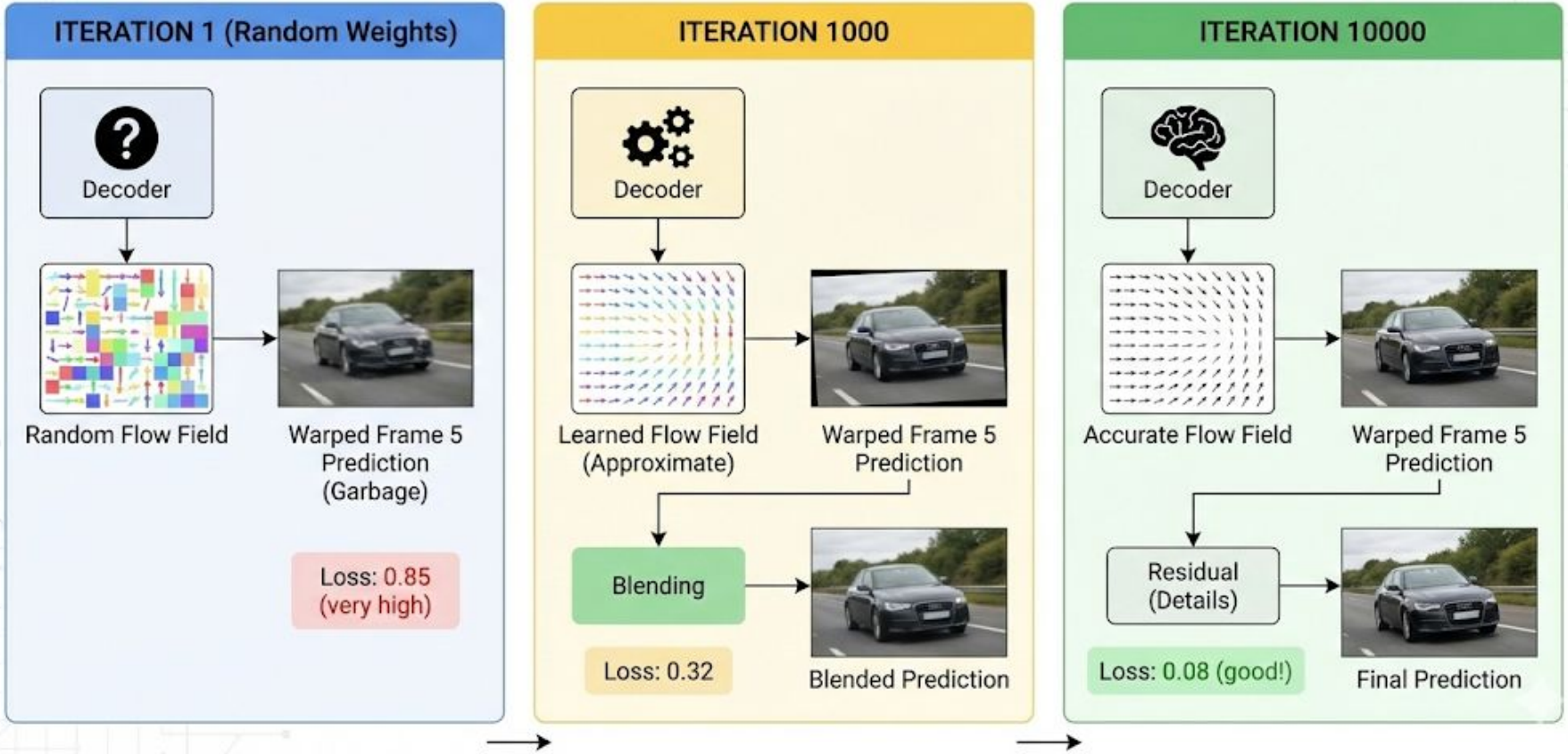
DECODER

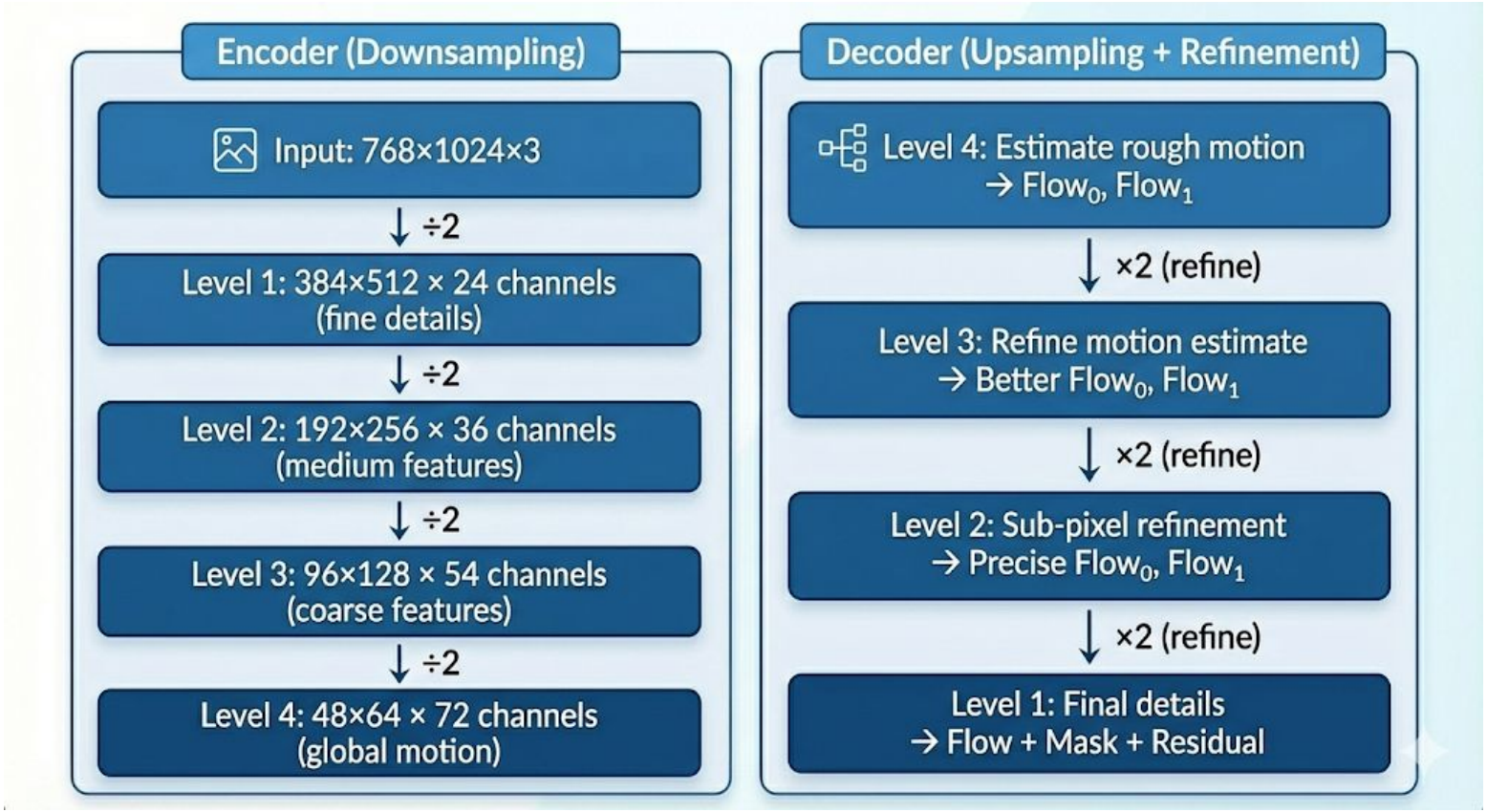
Estimated optical flow for frame 1(01) and 2(02)





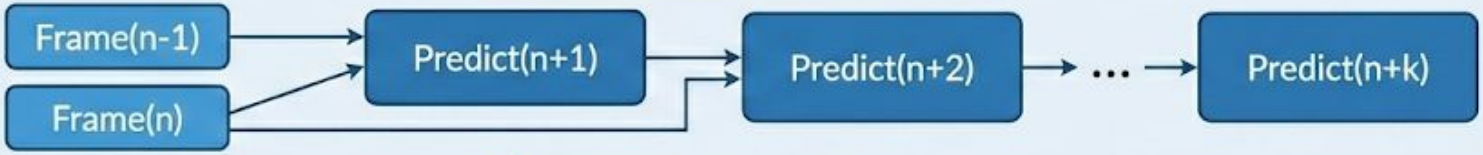
THE LEARNING PROCESS: FROM RANDOM NOISE TO ACCURATE PREDICTION




Encoder (Downsampling)
Input: 768×1024×3
÷2
Level 1: 384×512 × 24 channels
(fine details)
÷2
Level 2: 192×256 × 36 channels
(medium features)
÷2
Level 3: 96×128 × 54 channels
(coarse features)
÷2
Level 4: 48×64 × 72 channels
(global motion)
Decoder (Upsampling + Refinement)
Level 4: Estimate rough motion
→ Flow0, Flow1
×2 (refine)
Level 3: Refine motion estimate
→ Better Flow0, Flow1
×2 (refine)
Level 2: Sub-pixel refinement
→ Precise Flow0, Flow1
×2 (refine)
Level 1: Final details
→ Flow + Mask + Residual

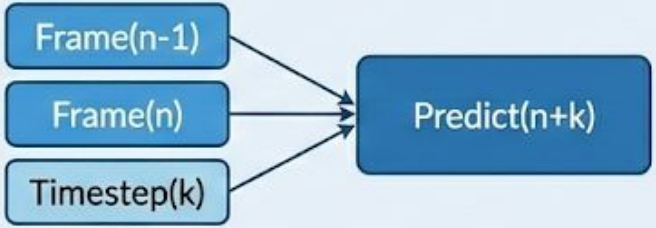
Three Approaches for Video Prediction

1. RECURRENT PREDICTION



⚠️ Error accumulates over steps

2. ARBITRARY PREDICTION

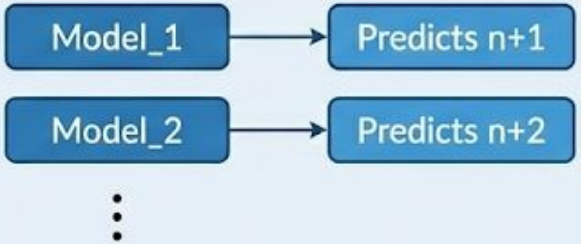


Single pass, timestep embedding controls prediction horizon

✅ No error accumulation

3. INDEPENDENT PREDICTION

Separate model trained for each prediction step

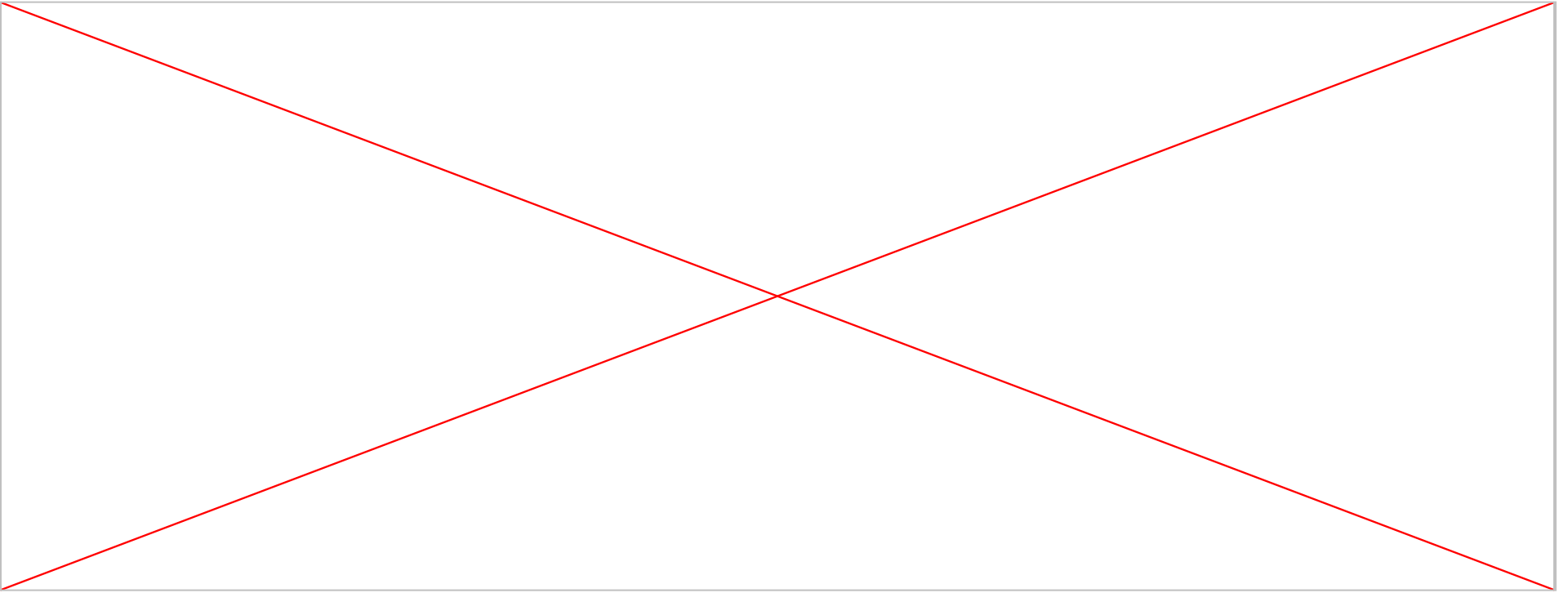


✅ ❌ Best accuracy, Multiple models needed

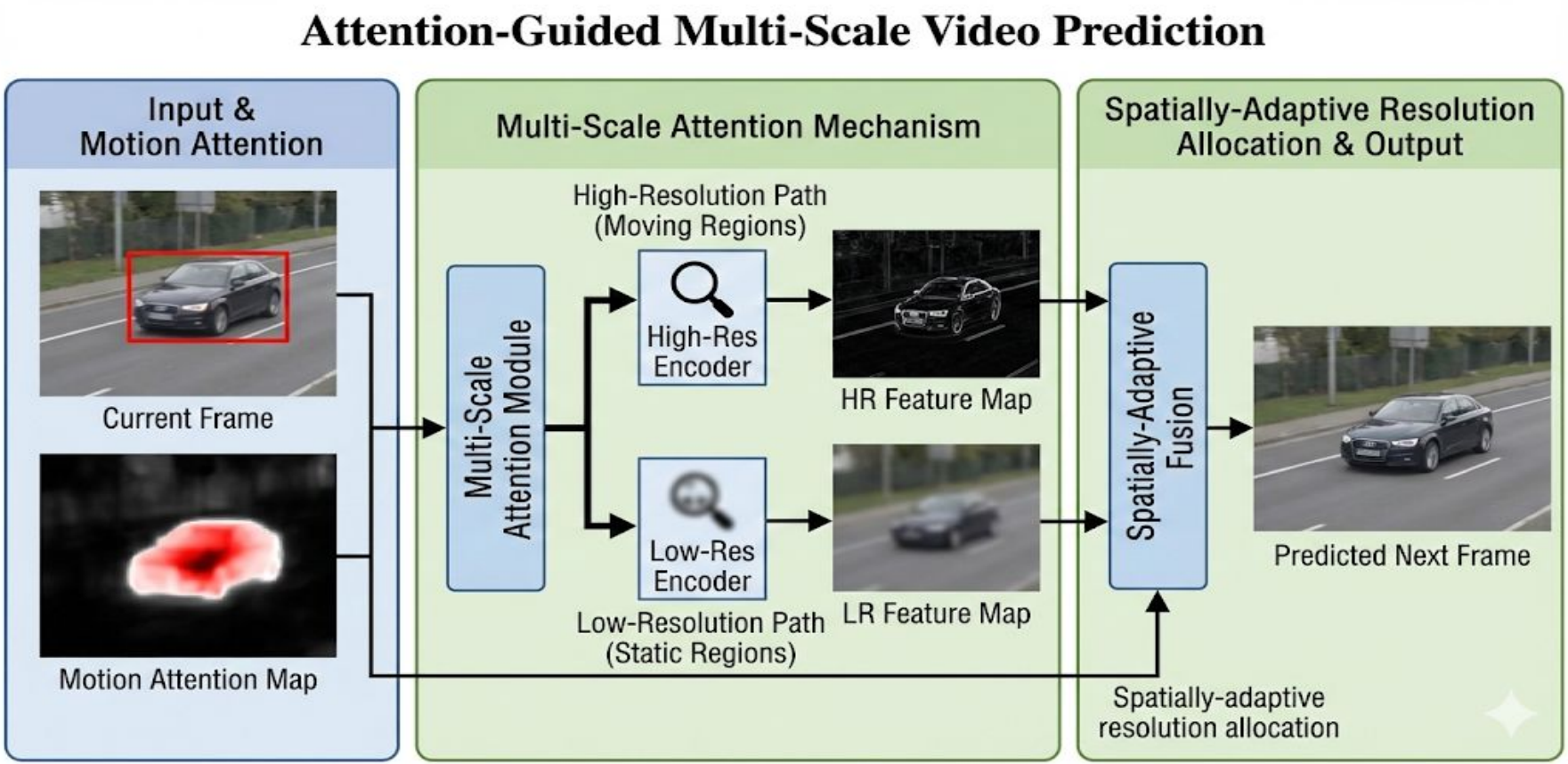
RESULTS

Frame size - 512x512
Fps - 15
PSNR - 33.08 db

When vehicles are going perpendicular

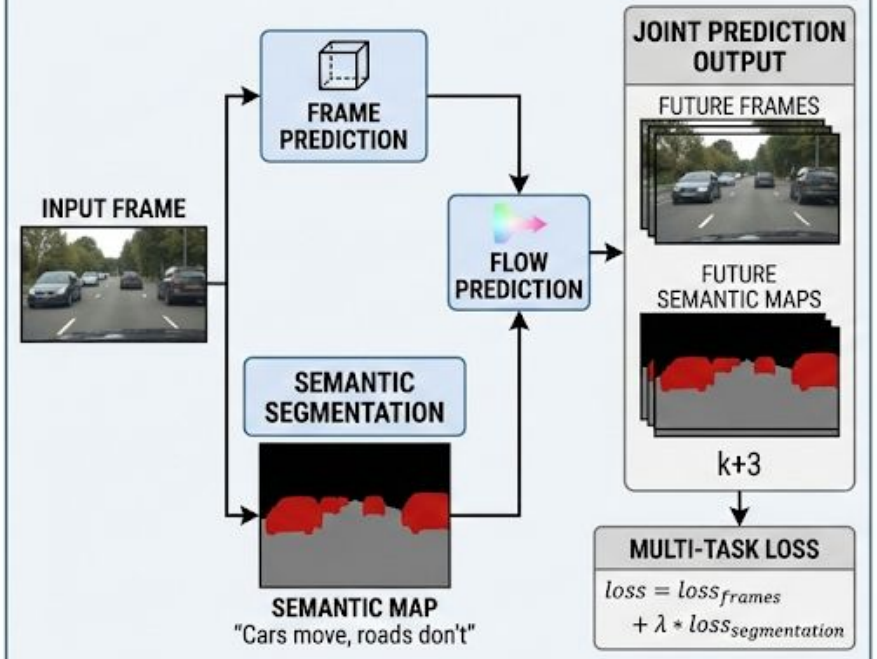


Future Work


Attention-Guided Multi-Scale Video Prediction
Input & Motion Attention
Current Frame
Motion Attention Map
Multi-Scale Attention Mechanism
Multi-Scale Attention Module
High-Resolution Path (Moving Regions)
High-Res Encoder
HR Feature Map
Low-Res Encoder
LR Feature Map
Low-Resolution Path (Static Regions)
Spatially-Adaptive Resolution Allocation & Output
Spatially-Adaptive Fusion
Predicted Next Frame
Spatially-adaptive resolution allocation

TWO APPROACHES TO ADVANCED VIDEO PREDICTION

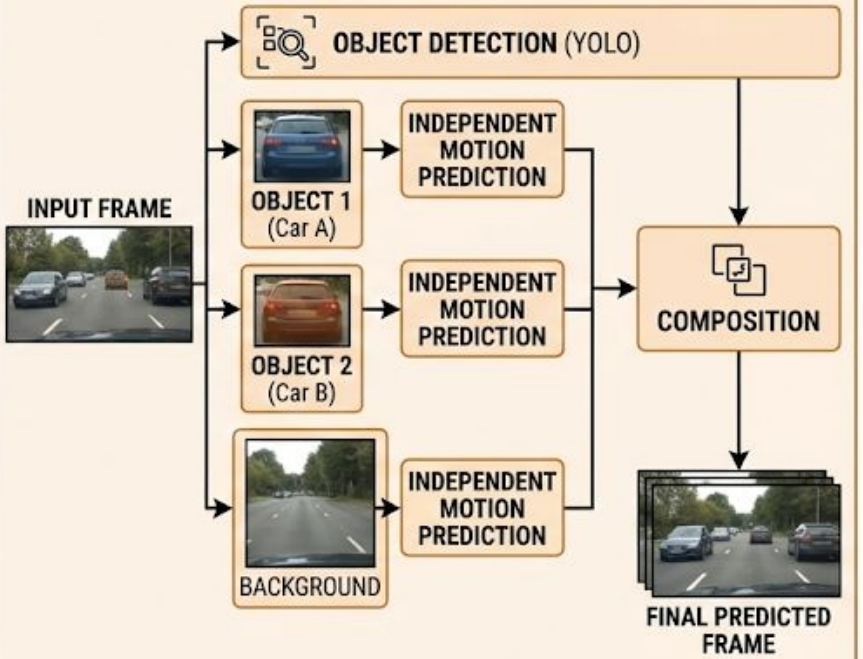
A) Semantic-Guided Prediction



NOVELTY First Joint Frame-Segmentation Prediction for k+3

Paper Angle: "Semantically-Informed Video Prediction"

B) Object-Centric Prediction



NOVELTY Compositional Approach to Video Prediction

Paper Angle: "Object-Compositional Video Prediction for Driving"